

# Super-Eddington Little Blue Dots Over-Reionize Helium

JWST's faint AGN may complete helium reionization too early, constraining super-Eddington accretion to under 10%.

arXiv:2608.28745 — Super-Eddington Little Blue Dots May Reionize Helium Too Early

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## Background

JWST found many faint AGN at  $z > 4$ , called Little Blue Dots, possibly super-Eddington. Their ionizing output could affect helium reionization timing.

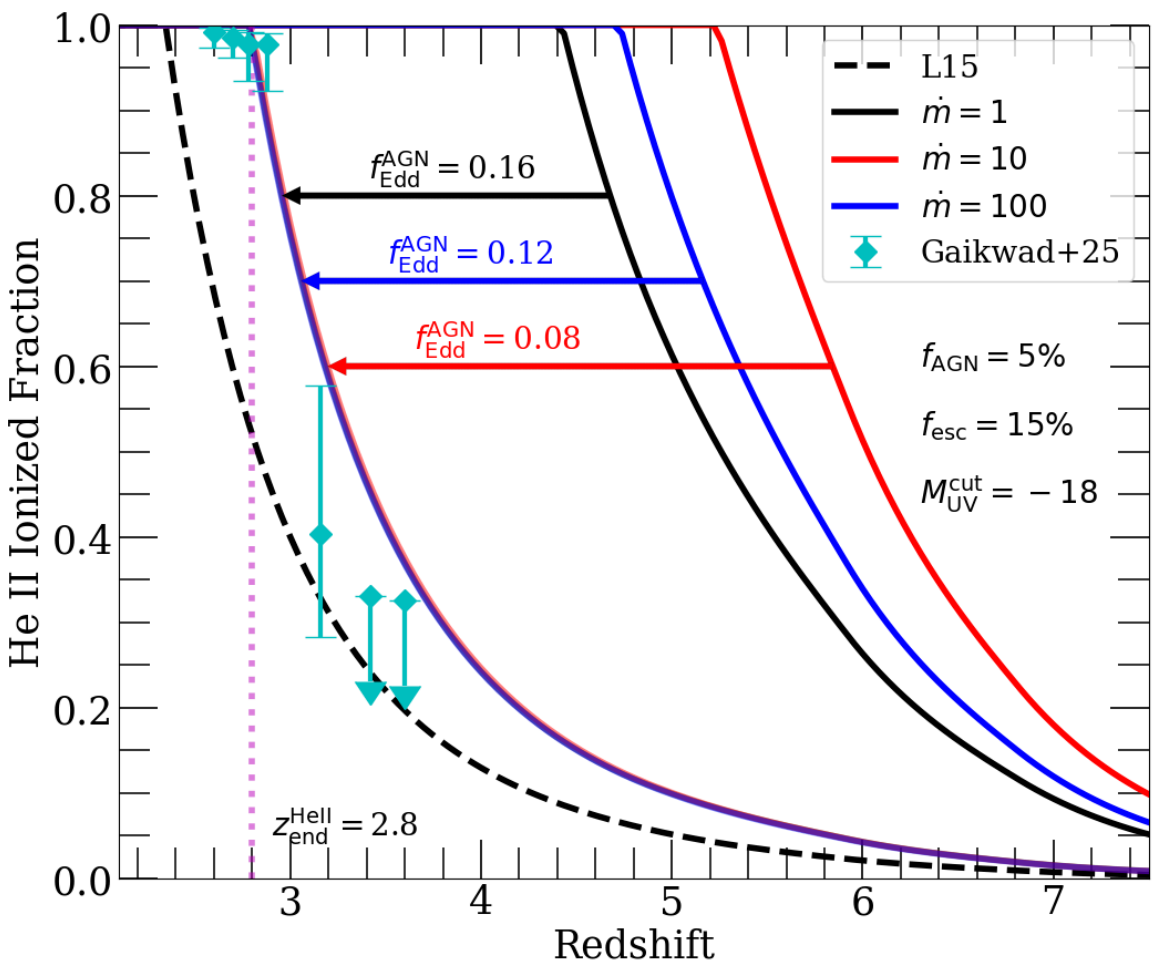
How does super-Eddington accretion in Little Blue Dots impact the timing of helium reionization?

## What this paper is selling

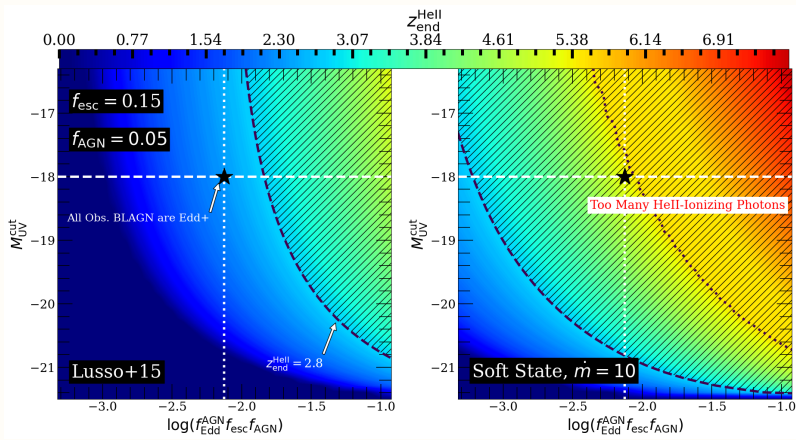
The paper shows that if most Little Blue Dots are super-Eddington, helium reionization ends too early, conflicting with observations. This limits the super-Eddington fraction to below 10%.

## Key results to remember

- Super-Eddington LBDs would end helium reionization before  $z=3$ , contradicting data.
- Only  $<10\%$  of LBDs can be super-Eddington under fiducial assumptions.
- Small escape fractions or lower abundance could allow more super-Eddington LBDs.



**Figure 3 |** Helium reionization histories for different SEDs; super-Eddington models complete reionization too early, conflicting with  $z=2.8$  constraints.



**Figure 4 |** Excluded AGN parameter space: hatched regions show where helium reionization ends too early, ruling out high super-Eddington fractions.